BA-BE Practice 3

Module 7a - Bioavailability and Bioequivalence · Daily practice sheet · No date

Name _____ Date _____

1. After oral administration of a single 1000 mg capsule of an investigational drug (half-life approximately 4.6 hours and apparent volume of distribution approximately 18 L) to a normal volunteer, the calculated AUC was 246 mg h/L.

a What is the bioavailability of this oral capsule in this volunteer?

b What is the expected AUC after administration of a single IV bolus dose of 500 mg to the same volunteer?

c What oral dose would provide equivalent bioavailability to the 500-mg IV bolus dose? (Round to the nearest ten) (The capsule is 0.667 bioavailable.)

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 0.667 (as a decimal fraction)

WHAT THE STEM GIVES

D po	1000 mg	the denominator of F
t½	4.6 hours	gives $k = \frac{0.693}{4.6} = 0.1507\ hr^{-1}$
V_D	18 L	$Cl_T = V_D \times k = 2.712$ L/hr
AUC po	246 mg h/L	multiplied by Cl_T: $2.712 \times 246 = 667$ mg absorbed

THE WORKING, ONE LINE AT A TIME

SETUP

$$F\ D\ po = Cl_T \times AUC\ po,\ so\ F = \frac{Cl_T \times AUC\ po}{D\ po},\ with\ Cl_T = k\ V_D$$

Clearance times the oral area is the amount that reached the circulation; no IV area is given, so the clearance comes from the half-life and the volume.

ALGEBRA

$$Cl_T = (18\ L)\left(\frac{0.693}{4.6\ hr}\right) = (18\ L)(0.1507\ hr^{-1}) = 2.712\ L/hr$$

Litres times reciprocal hours gives L/hr, the clearance that applies to the oral dose as well as to an IV dose.

ALGEBRA

$$F = \frac{(2.712\ L/hr)(246\ mg{\cdot}hr/L)}{1000\ mg} = \frac{667\ mg}{1000\ mg} = 0.667$$

L/hr × mg·hr/L leaves mg, the 667 mg that reached the circulation, and mg over mg is a pure fraction: two thirds of the capsule dose.

> **DOES THE NUMBER MAKE SENSE?**
> F lies between 0 and 1. Clearance 2.712 L/hr times the area 246 mg h/L is about 667 mg, two thirds of the 1000 mg capsule, so F near 0.667 fits; a value above 1 means the dose and the absorbed amount were swapped.

b

Her key

Answer: 184.4 (mg/L)hr

WHAT THE STEM GIVES

D po	1000 mg	*not needed: the IV area is asked*
t½	4.6 hours	gives $k = \frac{0.693}{4.6} = 0.1507\ hr^{-1}$
V_D	18 L	$Cl_T = V_D \times k = 2.712$ L/hr, the denominator
AUC po	246 mg h/L	*not needed: the IV area comes from dose and clearance*

D IV	500 mg	the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{D IV} = \text{Cl}_\text{T} \times \text{AUC IV}, \text{ so AUC IV} = \frac{\text{D IV}}{\text{Cl}_\text{T}}$$

An IV dose is wholly available, so the whole dose over the clearance is the area; the clearance is the same 2.712 L/hr as for the oral dose.

ALGEBRA

$$\text{AUC IV} = \frac{500 \text{ mg}}{(18 \text{ L})(0.1507 \text{ hr}^{-1})} = \frac{500 \text{ mg}}{2.712 \text{ L/hr}} = 184.4 \text{ (mg/L)hr}$$

mg over L/hr gives (mg/L)hr; the 500 mg IV dose gives a smaller area than the 1000 mg capsule, 184.4 against 246, because 667 mg of the capsule was absorbed.

> **DOES THE NUMBER MAKE SENSE?**
> 500 mg over a clearance of 2.712 L/hr: 2.712 × 184 is a little under 500, so about 184 (mg/L)hr; the IV area is below the capsule area of 246 because 500 mg is less than the two thirds of 1000 mg that was absorbed.

C Her key

Answer: 750 mg

WHAT THE STEM GIVES

D po	1000 mg	*not needed: the equivalent dose is asked*
t½	4.6 hours	*not needed: F is already given*
V_D	18 L	*not needed*
AUC po	246 mg h/L	*not needed*
D IV	500 mg	the numerator of D po = $\frac{\text{D IV}}{\text{F}}$
rounding	nearest ten	rounds 749.6 to 750 mg
F	0.667	the denominator

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{F} \times \text{D po} = \text{D IV}, \text{ so D po} = \frac{\text{D IV}}{\text{F}}$$

Equivalent bioavailability means the oral AUC equals the IV AUC, so the fraction absorbed times the oral dose must equal the whole 500 mg IV dose.

ALGEBRA

$$\text{D po} = \frac{500 \text{ mg}}{0.667} = 749.6 \text{ mg}$$

F has no units, so the oral dose is in milligrams; it is larger than 500 mg because only two thirds of the capsule dose reaches the circulation.

ROUND

D po = 750 mg

The stem asks for the nearest ten, so 749.6 mg is reported as 750 mg, a dose that can be given. Check: 0.667 × 750 mg = 500.3 mg reaches the circulation, matching the 500-mg IV bolus dose, so the capsule regimen gives the same AUC as the IV dose.

DOES THE NUMBER MAKE SENSE?

Two thirds of the capsule dose reaches the circulation, so the oral dose is half as much again as 500 mg: about 750 mg, and 749.6 rounds to 750. It must exceed 500 mg because F is below 1.